

DAFFODIL INTERNATIONAL UNIVERSITY



DEPARTMENT OF CIS

LAB REPORT -01

COURSE CODE : CIS 216

COURSE TITLE : OBJECT ORIENTED PROGRAMMING

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LAB REPORT

1.Code:

```
public class HelloWorld {  
    public static void main(String[] args)  
    {  
        System.out.println("Hello, World");  
    }  
}
```

Output:

Output

Hello, World

2.Code:

```
public class CommentsExample {  
    public static void main(String[] args) {  
        // This is a single-line comment  
        System.out.println("Hello, World"); // Comment at end of  
line  
  
        /*  
         * This is a multi-line comment  
         * It can span across multiple lines  
         */  
        int x = 10; // Variable declaration  
  
        /**  
         * This is a documentation comment (Javadoc)  
         * It is used to generate documentation for methods,  
classes, etc.  
         */  
        System.out.println("Value of x: " + x);  
    }  
}
```

Output:

Output

Hello, World
Value of x: 10

3.Code:

```
public class DataTypesDemo {
    public static void main(String[] args) {

        // ----- Primitive Data Types -----
        byte b = 100;           // 1 byte
        short s = 30000;        // 2 bytes
        int i = 100000;         // 4 bytes
        long l = 100000000000L; // 8 bytes

        float f = 3.14f;        // 4 bytes
        double d = 3.14159265359; // 8 bytes

        char c = 'A';           // 2 bytes (Unicode
character)
        boolean flag = true;    // 1 bit

        // ----- Non-Primitive Data Types -----
        String str = "Hello, Java"; // String (class in Java)
        int[] arr = {1, 2, 3, 4, 5}; // Array
        Integer wrapperInt = Integer.valueOf(50); // Wrapper
class example
        StringBuilder sb = new StringBuilder("Java"); // Class
obje

        System.out.println("byte: " + b);
        System.out.println("short: " + s);
        System.out.println("int: " + i);
        System.out.println("long: " + l);
        System.out.println("float: " + f);
        System.out.println("double: " + d);
    }
}
```

```
        System.out.println("char: " + c);
        System.out.println("boolean: " + flag);

        System.out.println("String: " + str);
        System.out.print("Array: ");
        for (int num : arr) {
            System.out.print(num + " ");
        }
        System.out.println();

        System.out.println("Wrapper Integer: " + wrapperInt);
        System.out.println("StringBuilder: " + sb);
    }
}
```

Output:

Output

```
byte: 100
short: 30000
int: 100000
long: 100000000000
float: 3.14
double: 3.14159265359
char: A
boolean: true
String: Hello, Java
Array: 1 2 3 4 5
Wrapper Integer: 50
StringBuilder: Java
```

4.Code:

```
public class Main {

    // Instance variable (belongs to each object)
    int instanceVar = 10;

    // Static variable (shared across all objects of the class)
    static String staticVar = "I am static";

    public void showVariables() {
        // Local variable (declared inside a method)
        int localVar = 5;

        System.out.println("Instance Variable: " + instanceVar);
        System.out.println("Static Variable: " + staticVar);
        System.out.println("Local Variable: " + localVar);
    }

    public static void main(String[] args) {
        // Creating object
        Main obj1 = new Main();
        obj1.showVariables();

        // Accessing static variable directly using class name
        System.out.println("Accessing Static Variable via class:
" + Main.staticVar);
    }
}
```

Output:

Output

```
Instance Variable: 10
Static Variable: I am static
Local Variable: 5
Accessing Static Variable via class: I am static
```

5.Code:

```
public class Main {
    public static void main(String[] args) {
```

```

int a = 10, b = 3;

// Arithmetic Operators
System.out.println("a + b = " + (a + b)); // Addition
System.out.println("a - b = " + (a - b)); // Subtraction

// Relational Operator
System.out.println("a > b ? " + (a > b)); // Greater than

// Logical Operator
boolean x = true, y = false;
System.out.println("x && y = " + (x && y)); // Logical AND

// Assignment Operator
a += 5; // a = a + 5
System.out.println("a after += 5 : " + a);

// Ternary Operator
int max = (a > b) ? a : b;
System.out.println("Maximum = " + max);
}
}

```

Output

```

a + b = 13
a - b = 7
a > b ? true
x && y = false
a after += 5 : 15
Maximum = 15

```

6.Code:

```

public class DecisionMakingDemo {
    public static void main(String[] args) {
        int number = 10;

        // if statement

```

```
if (number > 0) {
    System.out.println("The number is positive.");
}

// if-else statement
if (number % 2 == 0) {
    System.out.println("The number is even.");
} else {
    System.out.println("The number is odd.");
}

// if-else-if ladder
if (number < 0) {
    System.out.println("The number is negative.");
} else if (number == 0) {
    System.out.println("The number is zero.");
} else {
    System.out.println("The number is positive.");
}

// switch statement
int day = 3;
switch (day) {
    case 1:
        System.out.println("Monday");
        break;
    case 2:
        System.out.println("Tuesday");
        break;
    case 3:
        System.out.println("Wednesday");
        break;
    default:
        System.out.println("Other day");
}
}
```

Output:

Output

```
The number is positive.  
The number is even.  
The number is positive.  
Wednesday
```

7.Code:

```
public class LoopsDemo {  
    public static void main(String[] args) {  
  
        // 1. For loop  
        System.out.println("For Loop:");  
        for (int i = 1; i <= 5; i++) {  
            System.out.println("i = " + i);  
        }  
  
        // 2. While loop  
        System.out.println("\nWhile Loop:");  
        int j = 1;  
        while (j <= 5) {  
            System.out.println("j = " + j);  
            j++;  
        }  
  
        // 3. Do-While loop  
        System.out.println("\nDo-While Loop:");  
        int k = 1;  
        do {  
            System.out.println("k = " + k);  
            k++;  
        } while (k <= 5);  
  
        // 4. Enhanced For Loop (for-each loop)  
        System.out.println("\nEnhanced For Loop:");  
        int[] numbers = {10, 20, 30, 40, 50};  
        for (int num : numbers) {  
            System.out.println("num = " + num);  
        }  
    }  
}
```

```
}  
}
```

Output:

Output

While Loop:

```
j = 1  
j = 2  
j = 3  
j = 4  
j = 5
```

Do-While Loop:

```
k = 1  
...  
k = 2  
k = 3  
k = 4  
k = 5
```

Enhanced For Loop:

```
num = 10  
num = 20  
num = 30  
num = 40  
num = 50
```